General Project Documentation

CommandBrick – Assistive Bluetooth Keyboard and Mouse

Problem Statement

The CommandBrick project was developed as part of an assistive technology cocreation project for **Oliver**, a user with a **neuromuscular disease**. Due to his condition, Oliver has limited muscle strength and is unable to apply significant force when interacting with conventional computer peripherals such as keyboards or mechanical buttons.

Initially, Oliver requested a custom table that would allow him to securely position and use his existing Bluetooth keyboard. During discussions with the project team, it became clear that an even more accessible solution could be developed.

Instead of only designing a support table, the project evolved into the development of **CommandBrick**, an assistive Bluetooth input device that combines a capacitive touch keyboard and a touchpad into a single portable system. By replacing mechanical buttons with capacitive touch sensors, the device requires only a very light touch to activate commands, significantly reducing the physical effort required during computer use.

The goal of the project was therefore not only to fulfill the original request but also to improve accessibility and independence by providing a customized input device adapted to Oliver's physical abilities.

1. Project Overview

CommandBrick is an assistive Bluetooth Human Interface Device (HID) designed to provide an alternative method of computer interaction for users with limited hand mobility. The device combines a capacitive touch keyboard, a capacitive touchpad for mouse control, Bluetooth communication, and haptic feedback into a compact, battery-powered system based on the ESP32-S3 microcontroller.

Instead of using conventional mechanical buttons, CommandBrick relies on capacitive touch technology, allowing users to activate functions with minimal physical effort. By emulating a standard Bluetooth keyboard and mouse, the device can connect wirelessly to computers, tablets, and smartphones without requiring additional drivers.

The modular hardware and software architecture makes CommandBrick easy to reproduce, maintain, and extend for future accessibility applications.

2. Project Objectives

The primary objectives of the CommandBrick project were:

- Develop an assistive Bluetooth input device.
- Replace mechanical buttons with capacitive touch sensors.
- Integrate keyboard and mouse functionality into a single device.
- Provide tactile feedback using a vibration motor.
- Create a portable, rechargeable battery-powered system.
- Design a compact and ergonomic enclosure.
- Develop modular software that is easy to maintain and expand.

3. System Overview

CommandBrick consists of multiple hardware modules that work together under the control of the ESP32-S3 microcontroller.

The ESP32 continuously communicates with the Cirque Pinnacle TM040040 touchpad and the MPR121 capacitive touch controller through the I^2C interface. User interactions are processed in real time and converted into Bluetooth Human Interface Device (HID) reports, which are transmitted wirelessly to the connected computer.

Whenever a capacitive button is activated, the vibration motor provides tactile feedback to confirm that the user's input has been successfully detected.

The complete system is powered by a rechargeable Li-Ion battery, allowing fully portable operation.

4. Hardware Components

CommandBrick is built using the following hardware components:

Component	Purpose
ESP32-S3 Development Board	Main microcontroller
Cirque Pinnacle TM040040 Capacitive Touchpad	Mouse cursor control
MPR121 Capacitive Touch Controller	Capacitive button detection
Vibration Motor	Haptic feedback
3.7 V Li-Ion Battery	Portable power supply
TP4056 Charging & Protection Module	Battery charging and protection

5. Software Overview

The software was developed using the Arduino Framework for the ESP32-S3.

Its primary responsibilities include:

- Initializing all hardware components.
- Reading touchpad movement data.
- Detecting capacitive button presses.
- Processing user input.
- Sending Bluetooth keyboard and mouse commands.
- Controlling the vibration motor.

The ESP32-S3 operates as a Bluetooth Human Interface Device (HID), enabling the host operating system to recognize CommandBrick as a standard Bluetooth keyboard and mouse.

6. System Architecture

The interaction between the hardware components can be summarized as follows:

User Interaction

Cirque Pinnacle TM040040 Touchpad

MPR121 Capacitive Touch Controller

```
  │              │
    └──────────────────────────┬──────────────────────────┘
          │
      ESP32-S3
          │
    Bluetooth HID Reports
          │
    Computer / Tablet / Phone
          │
  Mouse & Keyboard Control
```

Whenever a capacitive button is pressed, the ESP32 also activates the vibration motor to provide tactile confirmation.

7. Communication Interfaces

CommandBrick uses two primary communication interfaces.

I²C Communication

The I²C bus connects the ESP32-S3 with:

- MPR121 Capacitive Touch Controller
- Cirque Pinnacle TM040040 Touchpad

Each device communicates using its own I²C address while sharing the same communication bus.

Bluetooth Low Energy (BLE)

Bluetooth Low Energy is used to transmit keyboard and mouse reports to the connected host using the Bluetooth HID profile.

Because CommandBrick behaves as a standard HID device, no additional software or drivers are required.

8. Device Functionality

CommandBrick combines multiple assistive input methods into a single device.

Mouse Control

The Cirque Pinnacle TM040040 touchpad detects finger movement and position. The ESP32 converts this information into Bluetooth mouse movement.

Keyboard Control

The MPR121 detects touches on twelve capacitive electrodes. Each electrode is assigned to a predefined keyboard or mouse function.

Haptic Feedback

Whenever a capacitive button is activated, the vibration motor briefly vibrates to provide tactile confirmation.

Portable Operation

The integrated rechargeable Li-Ion battery allows CommandBrick to operate wirelessly without an external power source.

9. Features

CommandBrick currently provides the following functionality:

- Bluetooth keyboard
- Bluetooth mouse
- Capacitive touch buttons
- Capacitive touchpad
- Haptic vibration feedback
- Rechargeable battery operation
- Compact hardware design
- Low-power operation
- Plug-and-play Bluetooth connectivity

10. Advantages

Compared to conventional assistive input devices, CommandBrick offers several advantages:

- Minimal activation force through capacitive touch technology.
- Wireless Bluetooth communication.
- Compact and lightweight construction.
- Combined keyboard and mouse functionality.
- Rechargeable battery-powered operation.
- Modular hardware and software architecture.
- Easy maintenance and future expansion.

11. Future Improvements

Possible future developments include:

- User-configurable button mappings.
- Adjustable cursor sensitivity.
- Multi-touch gesture support.
- Battery level monitoring.
- Over-the-Air (OTA) firmware updates.
- Multiple keyboard layouts.
- RGB status indicators.
- Companion configuration software.

12. Repository Contents

The GitHub repository contains all files required to understand, reproduce, and further develop the project.

The repository includes:

- ESP32-S3 source code
- Programming documentation
- Hardware documentation
- Pin documentation
- Wiring diagrams
- Device photographs
- Datasheets
- Project presentation
- Additional technical documentation